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SENIOR FIVE TERM 1

TOPIC 3/6: Coordinate Geometry 1

Competency: The learner analyses linear relationships to solve geometric problems and interpret spatial relationships in real-life situations

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Coordinate geometry 1

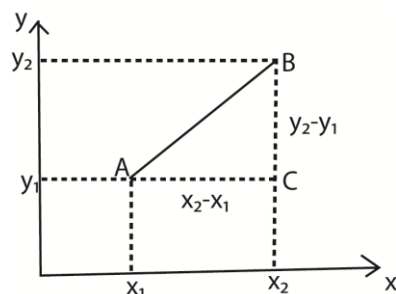
This is the area of mathematics where geometrical relationships are described algebraically by reference to the coordinates

The length of the line segment

Given two points $A(x_1, y_1)$ and $B(x_2, y_2)$ in x-y plane, the distance between A and B, denoted by $\overline{AB}$ is $\overline{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Proof

Geometrical approach



Using Pythagoras theorem

$$\overline{AB}^2 = \overline{AC}^2 + \overline{BC}^2$$

$$\overline{AB}^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\overline{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Vector method approach

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA}$$

$$= \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} - \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$= \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Using Pythagoras theorem

$$\overline{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example 1

(a) Find the distance between the points

(i) $A(1, 3)$ and $B(7, 11)$

Solution

$$\text{Using } \overline{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\overline{AB} = \sqrt{(7 - 1)^2 + (11 - 3)^2}$$

$$= \sqrt{6^2 + 8^2}$$

$$= \sqrt{100} = 10$$

(ii) $P(-1, 2)$ and $Q(3, 7)$

Solution

$$\overline{PQ} = \sqrt{(-1 - 3)^2 + (2 - 7)^2}$$

$$= \sqrt{(-4)^2 + (-5)^2}$$

$$= \sqrt{25} = 5$$

(b) The points A, B and C have coordinates $A(-3, 2)$, $B(-1, -2)$ and $C(0, n)$ where n is a constant. Given that $\overline{BC} = \frac{1}{5}\overline{AC}$, find the possible values of n.

Solution

$$\text{Using } \overline{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\overline{BC} = \sqrt{(0 - (-1))^2 + (n - (-2))^2}$$

$$= \sqrt{1^2 + (n + 2)^2}$$

$$= \sqrt{n^2 + 4n + 5}$$

$$\overline{AC} = \sqrt{(0 - (-3))^2 + (n - 2)^2}$$

$$= \sqrt{3^2 + (n - 2)^2}$$

$$= \sqrt{n^2 - 4n + 13}$$

$$\text{But } \overline{BC} = \frac{1}{5}\overline{AC}$$

$$\Rightarrow \sqrt{n^2 + 4n + 5} = \frac{1}{5}\sqrt{n^2 - 4n + 13}$$

$$5\sqrt{n^2 + 4n + 5} = \sqrt{n^2 - 4n + 13}$$

Squaring both sides

$$25(n^2 + 4n + 5) = n^2 - 4n + 13$$

$$24n^2 + 104n + 112 = 0$$

$$3n^2 + 13n + 14 = 0$$

$$(3n + 7)(n + 2) = 0$$

$$\text{Either } n = -\frac{7}{3} \text{ or } n = -2$$

Hence the values of n are $-\frac{2}{3}$ and -2

To show that given points are vertices of a right-angle triangle.

Suppose that the points A, B and C are vertices of a triangle ABC, to show that ABC is a right angled triangle, then by applying the Pythagoras theorem

$$\text{Either } \overline{AB}^2 = \overline{AC}^2 + \overline{BC}^2$$

$$\text{or } \overline{BC}^2 = \overline{AB}^2 + \overline{AC}^2$$

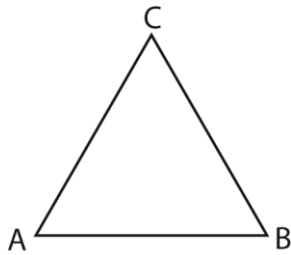
$$\text{or } \overline{AC}^2 = \overline{AB}^2 + \overline{BC}^2$$

Example 2

Prove that the following points are vertices of a right-angled triangle

- (a) A(2, 3), B(5, 6) and C(8, 3)

Solution



$$\overline{AC}^2 = (8 - 2)^2 + (3 - 3)^2 = 36$$

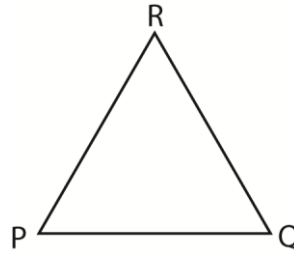
$$\overline{AB}^2 = (5 - 2)^2 + (6 - 3)^2 = 18$$

$$\overline{BC}^2 = (8 - 5)^2 + (3 - 6)^2 = 18$$

Since $\overline{AB}^2 + \overline{BC}^2 = \overline{AC}^2 = 36$, the triangle ABC is a right angled triangle

- (b) P(2, 1), Q(5, -1) and R(9, 5)

Solution



$$\overline{PQ}^2 = (5 - 2)^2 + (-1 - 1)^2 = 13$$

$$\overline{PR}^2 = (9 - 2)^2 + (5 - 1)^2 = 65$$

$$\overline{QR}^2 = (9 - 5)^2 + (5 - (-1))^2 = 52$$

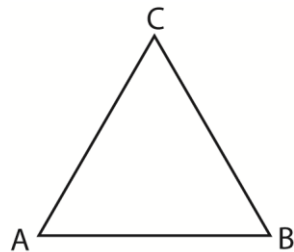
Since $\overline{PQ}^2 + \overline{QR}^2 = \overline{PR}^2 = 65$, the triangle PQR is a right angled triangle

Note: if the triangle is **isosceles**, then two of the sides must be equal and for **equilateral** triangle all the sides must be equal

Example 3

Prove that the following points A(1, 2), B(3, 7) and C(6, 14) are vertices of an isosceles triangle.

Solution



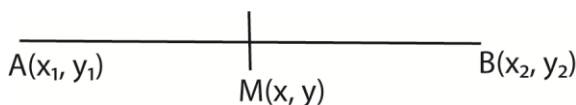
$$\overline{AC}^2 = (6 - 1)^2 + (14 - 2)^2 = 169$$

$$\overline{AB}^2 = (3 - 1)^2 + (7 - 2)^2 = 169$$

$$\overline{BC}^2 = (6 - 3)^2 + (14 - 7)^2 = 98$$

Since $\overline{AC}^2 = \overline{AB}^2 = 169$, hence, the triangle ABC, is isosceles triangle.

Finding the mid-point of a line segment



The mid-point, M of a line segment AB with A(x₁, y₁) and B(x₂, y₂) is given as

$$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

Example 4

- (a) Find the coordinates of the midpoint of the line joining each of the following pairs of points

- (i) A(8, 4) and B(2, -4)

Solution

$$\begin{aligned}\text{The midpoint of AB} &= M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right) \\ &= M\left(\frac{8+2}{2}, \frac{4-4}{2}\right) \\ &= M(5, 0)\end{aligned}$$

- (ii) P(-6, -2) and Q(-4, -5)

Solution

$$\begin{aligned}\text{The midpoint of PQ} &= M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right) \\ &= M\left(\frac{-6-4}{2}, \frac{-2-5}{2}\right) \\ &= M(-5, -3.5)\end{aligned}$$

- (b) Find the coordinates of point S given that M(3, -2) is the midpoint of the straight line joining S to T(9, -2)

Solution

$$\text{The midpoint of PQ} = M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$\begin{aligned}\Rightarrow \frac{x+9}{2} &= 3 \\ x &= -3\end{aligned}$$

$$\text{Also } \frac{y-2}{2} = -2$$

$$y = -2$$

Hence S(-3, -2)

Revision Exercise 1

1. Find the distance between each of the following pairs of points

- (a) (-7, 3) and (-2, 5) $[\sqrt{29}]$

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- (b) (2, -3) and (7, 7) $[5\sqrt{5}]$
(c) (4, -1) and (-2, 1) $[2\sqrt{10}]$

2. Prove that a triangle with vertices (1, 2), (13, 7) and (6, 14) is isosceles.
3. Find the midpoint of the following point
 - (a) (2, 1) and (4, 5) $[3, 3]$
 - (b) (-1, 4) and (3, 1) $[1, 3]$
 - (c) (-2, 6) and (0, 2) $[-1, 4]$
4. Prove that the points A(-2, 0), B(0, $2\sqrt{5}$) and C(2, 0) are vertices of an equilateral triangle.
5. The points L, M and N have coordinates (3, 1), (2, 6) and (x, 5) respectively. Given that the distance LM is equal to the distance MN, calculate the possible values of x. $[-3 \text{ or } 7]$
6. Given that the distance between P(r, 4) and Q(2, 3) is equal to the distance between R(3, -1) and S(-2, 4). Calculate the possible value of r. $[-5 \text{ or } 7]$
7. A triangle has vertices A(6, 2), B(x, 6) and C(-2, 6). Given that the triangle is isosceles with AB = BC, Calculate the value of x. $[3]$
8. F(5, 1), G(x, 7) and H(8, 2) are vertices of a triangle. Given that the length of the side FG is twice the length of side FH, find the value of x. $[3 \text{ or } 7]$
9. Given that the distance from A(13, 10) to B(1, y) is three times the distance from B to C(-3, -2), find the value of y. $[1 \text{ or } -8]$
10. M(6, 5) is the midpoint of a straight line joining the point A to point B, find the coordinates of B $[10, 7]$

Calculating the gradient of a straight line

The gradient of a line joining points A(x₁, y₁) and B(x₂, y₂) is the measure of steepness of the line AB and it is a ratio of the change in y-coordinate to the change in x-coordinate.

$$\text{i.e., gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

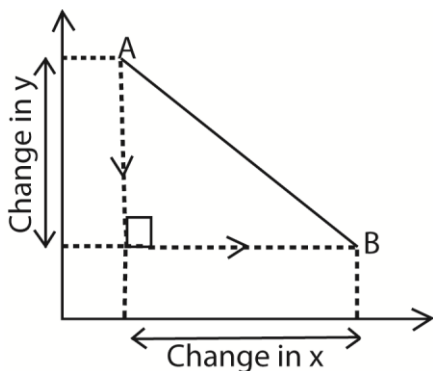
The gradient is usually denoted by m which may be positive or negative

$$= \frac{y_2 - y_1}{x_2 - x_1}$$

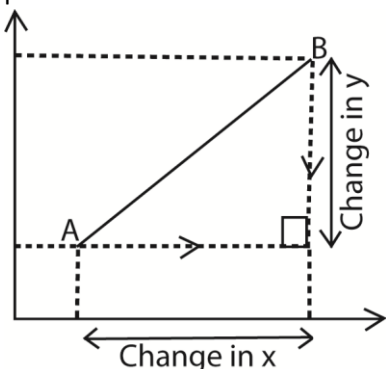
$$\theta = \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right)$$

Note:

- (i) If the line slopes downwards from left to right, the gradient

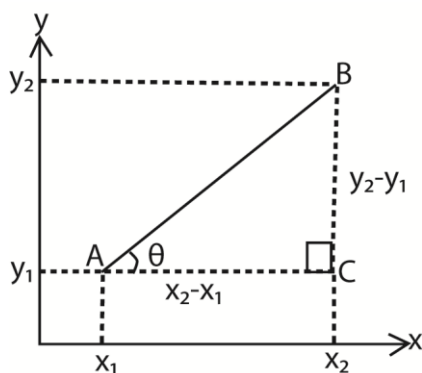


- (ii) On the other hand, if the line slopes upwards from left to right, the gradient is positive.



Angle of a straight line to the horizontal

Suppose that the line in the second illustration makes θ is the horizontal as shown below



From trigonometry, $\tan \theta = \frac{BC}{AC}$

This means that if $m = \frac{y_2 - y_1}{x_2 - x_1}$

$$\theta = \tan^{-1} m$$

Example 5

- (a) Find the gradient of the straight line joining each of the following pairs of points.

- (i) A(7, 4) and B(-1, -2)

Solution

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 - 4}{-1 - 7} = \frac{-6}{-8} = \frac{3}{4}$$

- (ii) A(-3, -2) and B(-4, -5)

Solution

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-5 - (-2)}{-4 - (-3)} = \frac{-3}{-1} = 3$$

- (b) Find the angle which the straight line joining each of the following pairs of points makes with the horizontal

- (i) A(5, 4) and B(6, 8)

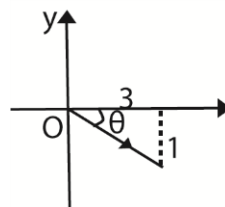
Solution

$$\begin{aligned} \text{Angle } \theta &= \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) \\ &= \tan^{-1} \left(\frac{8 - 4}{6 - 5} \right) \\ &= 76^\circ \end{aligned}$$

- (ii) A(-3, -5) and (-4, -2)

$$\begin{aligned} \text{Angle } \theta &= \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) \\ &= \tan^{-1} \left(\frac{-2 - (-5)}{-4 - (-3)} \right) \\ &= -71.6^\circ \end{aligned}$$

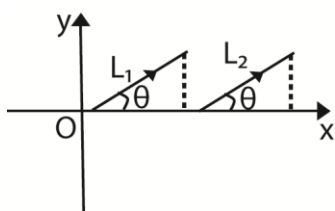
i.e.



Hence the angle which AB makes with the horizontal is 71.6° with positive x-axis downwards as shown in the diagram.

Calculating gradient of parallel lines

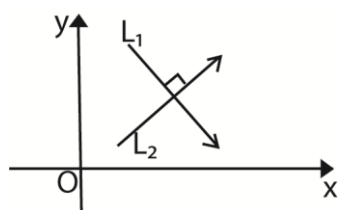
If two are parallel, their gradients are equal.



In the diagram above L_1 is parallel to L_2 and both L_1 and L_2 have the same gradient.

Gradient of perpendicular lines

If two lines are perpendicular, the product of their gradients is -1.



If m_1 and m_2 are the gradients of L_1 and L_2 respectively, then $m_1 \times m_2 = -1$

Example 6

- (a) Given the points A(2, 3), B(5, 5), C(7, 2) and D(4, 0)

- (i) Prove that AB is parallel to DC

$$\text{Gradient of AB, } m_1 = \frac{5-3}{5-2} = \frac{2}{3}$$

$$\text{Gradient of DC, } m_2 = \frac{2-0}{7-4} = \frac{2}{3}$$

Since the gradient of AB and DC are equal, the lines AB and DC are parallel

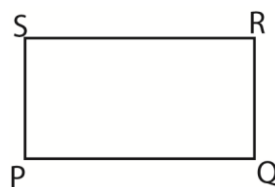
- (ii) Prove that AC is perpendicular to BD

$$\text{Gradient of AC, } m_1 = \frac{2-3}{7-2} = \frac{-1}{5}$$

$$\text{Gradient of BD, } m_2 = \frac{0-5}{4-5} = 5$$

Since $m_1 \times m_2 = -1$, AC is perpendicular to BD

- (b) Prove that the point P(1, 3), Q(3, 4), R(5, 0) and S(3, -1) form a parallelogram.



$$\text{Gradient of PQ, } m_1 = \frac{4-3}{3-1} = \frac{1}{2}$$

$$\text{Gradient of SR, } m_2 = \frac{0-(-1)}{5-3} = \frac{1}{2}$$

$\therefore$ PQ and SR are parallel

$$\overline{PQ} = \sqrt{(3-1)^2 + (4-3)^2} = \sqrt{5}$$

$$\overline{SR} = \sqrt{(5-3)^2 + (0-(-1))^2} = \sqrt{5}$$

$\therefore$ PQ and SR are equal

$$\text{Gradient of PS, } m_1 = \frac{-1-2}{3-1} = -2$$

$$\text{Gradient of QR, } m_2 = \frac{0-4}{5-3} = -2$$

$\therefore$ PS and QR are parallel

So the figure PQRS could either be a rectangle or a parallelogram

For a rectangle

PQ and PS are perpendicular, thus the product of their gradients = -1

$$\text{Since the gradient of PQ and } \times \text{ gradient of PS} = \frac{1}{2} \times -2 = -1$$

Hence the figure PQRS is a rectangle not a parallelogram.

- (c) The quadrilateral ABCD has vertices A(-2, -3), B(1, -1), C(7, -10) and D(2, -9).

- (i) Prove that AD is parallel to BC

$$\text{Gradient of AD, } m_1 = \frac{-9-(-3)}{2-(-2)} = \frac{-3}{2}$$

$$\text{Gradient of BC, } m_2 = \frac{-10-(-1)}{7-1} = \frac{-3}{2}$$

Since the gradient of AD and BC are equal, the lines AD and BC are parallel

- (ii) Prove that AD is perpendicular to BC

$$\text{Gradient of AD, } m_1 = \frac{-1-(-3)}{1-(-2)} = \frac{2}{3}$$

$$\text{Gradient of BC, } m_2 = \frac{-10-(-1)}{7-1} = \frac{-3}{2}$$

Since $m_1 \times m_2 = -1$, AD is perpendicular BC

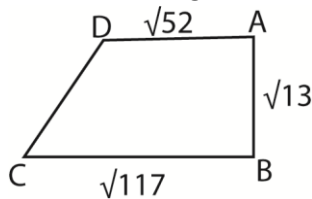
- (iii) Prove that the area of the quadrilateral ABCD is $32\frac{1}{2}$ sq. units

$$\overline{AB} = \sqrt{(1 - (-2))^2 + (-1 - (-3))^2} \\ = \sqrt{13}$$

$$\overline{BC} = \sqrt{(7 - 1)^2 + (-10 - (-1))^2} \\ = \sqrt{117}$$

$$\overline{AD} = \sqrt{(2 - (-2))^2 + (-9 - (-3))^2} \\ = \sqrt{52}$$

Since all the sides of a quadrilateral are different, the figure is a trapezium



$$\text{Area} = \frac{1}{2} \sqrt{13} (\sqrt{52} + \sqrt{117}) \\ = 32\frac{1}{2} \text{ sq. units}$$

- (d) A quadrilateral ABCD has vertices A(-2, 1), B(0, 4), C(3, 2) and D(1, -1)

- (i) Prove that all sides of the quadrilateral have the same length.

Solution

$$\overline{AB} = \sqrt{(0 - (-2))^2 + (4 - 1)^2} \\ = \sqrt{13}$$

$$\overline{BC} = \sqrt{(3 - 0)^2 + (2 - 4)^2} \\ = \sqrt{13}$$

$$\overline{CD} = \sqrt{(1 - 3)^2 + (-1 - 2)^2} \\ = \sqrt{13}$$

$$\overline{AD} = \sqrt{(-2 - 1)^2 + (1 - (-2))^2} \\ = \sqrt{13}$$

Hence all the sides of the quadrilateral are equal in length

- (ii) Prove that AB is parallel to AD

$$\text{Gradient of AB, } m_1 = \frac{4-1}{0-(-2)} = \frac{3}{2}$$

$$\text{Gradient of DC, } m_2 = \frac{2-(-1)}{3-1} = \frac{3}{2}$$

Since the gradient of AB and DC are equal, the lines AB and DC are parallel

- (iii) Prove that AD is parallel to BC

$$\text{Gradient of AD, } m_1 = \frac{-1-1}{1-(-2)} = \frac{-2}{3}$$

$$\text{Gradient of BC, } m_2 = \frac{2-4}{3-0} = \frac{-2}{3}$$

Since the gradients of AD and BC are equal, the lines AD and BC are parallel

- (iv) What is the name of the quadrilateral ABCD?

It could be a square or a rhombus

For square, gradient of AB $\times$ gradient of

$$BC = \frac{-2}{3} \times \frac{3}{2} = -1$$

Hence the quadrilateral is a square.

Revision Exercise 2

1. Find the gradient of the straight line of each of the following pairs of points.

(a) (-2, 5) and (5, -3) $\left[\frac{-8}{7} \right]$

(b) (3, 7) and (7, -4) $\left[\frac{-11}{4} \right]$

(c) (6, 3) and (7, 4) $[1]$

2. Find the angle between a line joining the following points with the horizontal

(a) (2, 5) and (-3, -2) $[54.46^\circ]$

(b) (3, 7) and (-6, 11) $[-23.96^\circ]$

(c) (5, -3) and (5, 2) $[90^\circ]$

3. A triangle has vertices A(3, -2), B(2, -14) and C(-2, -4). Find the gradients of the straight lines AB, BC and CA. Hence prove that the triangle is right-angled

$\left[12, \frac{-5}{2}, \frac{2}{5}; \text{hence BC and CA are perpendicular} \right]$

4. The straight line joining the points P(6, 5) to Q(q, 2) is perpendicular to the straight line joining point Q to R(9, -1). Find the value of q. [3, 11]
5. Prove that the quadrilateral PQRS with vertices P(-1, 3), Q(2, 4), R(4, -2) and S(1, -3) is a rectangle and calculate the area [20 sq. units]
6. The four points A(5, 4), B(6, 2), C(12, 5) and D(11, 7) are vertices of a quadrilateral. Prove that the quadrilateral is a rectangle and calculate its area. [15 sq. units]
7. Prove that the points A(2, 3), B(4, 8), C(8, 9) and D(4, -1) form a trapezium.
8. The quadrilateral ABCD has vertices S(1, 1), T(4, 5), U(12, -1) and V(1, -1) are vertices of a quadrilateral STUV.
 - (a) Prove that ST is perpendicular to TU, and that SV is perpendicular to UV.
 - (b) Calculate the length of each of the sides ST, TU, UV, and VS. [5, 10, 11, 2]
 - (c) Prove that the area of the quadrilateral STUV is 36 square units.
9. The quadrilateral CDEF has vertices, C(4, 0), D(8, 4), E(2, -8) and F(0, 2). The points P, Q, R and S are midpoints of the sides CD, DE, EF and FC respectively. Prove that the Quadrilateral PQRS is a rhombus and show that its area is 15 sq. units.

Equation of a straight line

The general equation of a straight line is given by $y = mx + c$, where m = gradient of the line i.e. $m = \frac{y_2 - y_1}{x_2 - x_1}$ and c = y-intercept. For lines passing through the origin, $c = 0$; hence $y = mx$.

Example 7

- (a) Find the equation of with gradient = 5 and passes through the point
- (i) A(2, 5)

Solution

Method 1

The general equation of a line is $y = mx + c$

Substituting for $m = 5$ and points of A

$$5 = 5(2) + c$$

$$c = -5$$

Hence the line is $y = 5x - 5$

Method 2

Let B(x, y) lie on the line

$$\text{From } m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$5 = \frac{y - 5}{x - 2}$$

$$y - 5 = 5(x - 2)$$

$$y = 5x - 5$$

- (ii) P(2, 7)

Using $y = mx + c$

Substituting for $m = 5$ and points of P

$$7 = 5(2) + c$$

$$c = 17$$

Hence the line is $y = 5x + 17$

- (b) Find the equation of a straight line joining the following points

- (i) A(0, 5) and B(3, 4)

Solution

$$\text{Gradient} = \frac{4 - 5}{3 - 0} = \frac{-1}{3}$$

Substituting coordinates for A in general equation

$$5 = \frac{-1}{3}(0) + c \Rightarrow c = 5$$

Hence equation of the line is $y = \frac{-1}{3}x + 5$

- (ii) P(-2, -5) and Q(3, -7)

Solution

$$\text{Gradient} = \frac{-7 - (-5)}{3 - (-2)} = \frac{-2}{5}$$

Substituting coordinates for P in general equation

$$-5 = \frac{-2}{5}(-2) + c \Rightarrow c = -5 - \frac{4}{5} = -\frac{29}{5}$$

Hence equation of the line is $y = \frac{-2}{5}x - \frac{29}{5}$

- (c) Find the equation of a perpendicular bisector of the line joining the following points

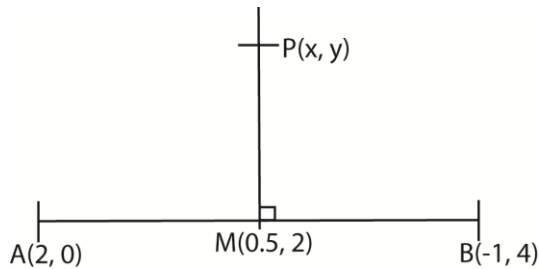
- (i) (2, 0) and (-1, 4)

Solution

Let M be the midpoint of AB

$$\Rightarrow M\left(\frac{2-1}{2}, \frac{0+4}{2}\right) = M(0.5, 2)$$

Let $P(x, y)$ lie on the perpendicular bisector



$$\text{Gradient of AB, } m_1 = \frac{4-0}{-1-2} = \frac{4}{-3}$$

$$\text{Gradient of MP, } m_2 = \frac{y-2}{x-0.5}$$

$$\text{But } m_1 \times m_2 = -1$$

$$\frac{4}{-3} \left(\frac{y-2}{x-0.5} \right) = -1$$

$$4(y-2) = 3\left(x - \frac{1}{2}\right)$$

$$8y - 16 = 6x - 3$$

$$8y - 6x = 13$$

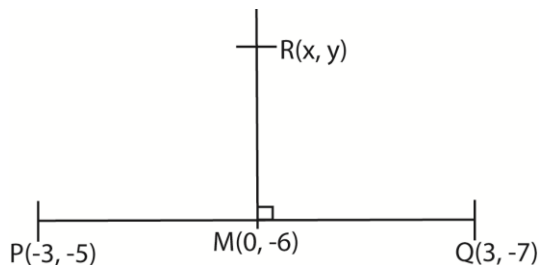
- (ii) $P(-3, -5)$ and $Q(3, -7)$

Solution

Let M be the midpoint of AB

$$\Rightarrow M\left(\frac{-3+3}{2}, \frac{-5-7}{2}\right) = M(0, -6)$$

Let $R(x, y)$ lie on the perpendicular bisector



$$\text{Gradient of PQ, } m_1 = \frac{8-(-7)}{-3-3} = \frac{15}{-6} = \frac{5}{-2}$$

$$\text{Gradient of MR, } m_2 = \frac{y-(-6)}{x-0} = \frac{y+6}{x-0}$$

$$\text{But } m_1 \times m_2 = -1$$

$$\frac{5}{-2} \left(\frac{y+6}{x-0} \right) = -1$$

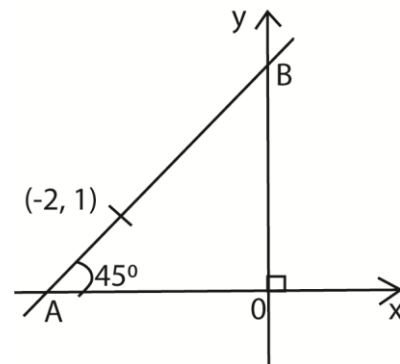
$$5(y+6) = 2x$$

$$5y - 2x + 30 = 0$$

- (d) A straight line, L passes through point $(-2, 1)$ and makes an angle of 45° with the horizontal.

- (i) Find the equation of the line L .

Solution



$$\text{Gradient of the line } L = \tan 45^\circ = 1$$

$$\Rightarrow m = 1$$

Substituting the coordinates of the point in the general equation, $y = mx + c$

$$1 = 1(-2) + c \Rightarrow \text{that } c = 3$$

Hence the equation of the line L is $y = x + 3$

- (ii) Given that L intersects the x -axis at A and the y -axis at B find the distance AB

Solution

At A , $y = 0$; $x = -3$

Hence coordinates of $A(-3, 0)$

At B $x = 0$ $y = 3$

Hence coordinates of $B(0, 3)$

$$\begin{aligned} \overline{AB} &= \sqrt{(0 - (-3))^2 + (3 - 0)^2} \\ &= 3\sqrt{2} \text{ units} \end{aligned}$$

Perpendicular distance of a point to a line

Just like in vectors, the shortest distance of a point from a given line is the perpendicular distance of a point from the line

Suppose that the equation of the line is in the form $ax + by + c = 0$

The perpendicular distance of a point

$$P(x_1, y_1) = d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

Example 8

Find the perpendicular distance of the following points from the given lines

- (a) $(-2, 6)$ from $x + y + 4 = 0$

Solution

$$a = 1, b = 1 \text{ and } c = 4$$

Substituting for $(x, y) = (-2, 6)$

$$d = \left| \frac{1(-2) + 1(6) + 4}{\sqrt{1^2 + 1^2}} \right| = \frac{8}{\sqrt{2}} = 4\sqrt{2} \text{ units}$$

- (b) $(7, -4)$ from $3x - 5y = 7$

Solution

Rearranging the equation

$$3x - 5y - 7 = 0$$

$$a = 3, b = -5 \text{ and } c = -7$$

Substituting for $(x, y) = (7, -4)$

$$d = \left| \frac{3(7) - 5(-4) - 7}{\sqrt{3^2 + (-5)^2}} \right| = \frac{34}{\sqrt{34}} = \sqrt{34} \text{ units}$$

- (c) $(8, -5)$ from $y = 3x - 1$

Solution

Rearranging the equation

$$3x - y - 1 = 0$$

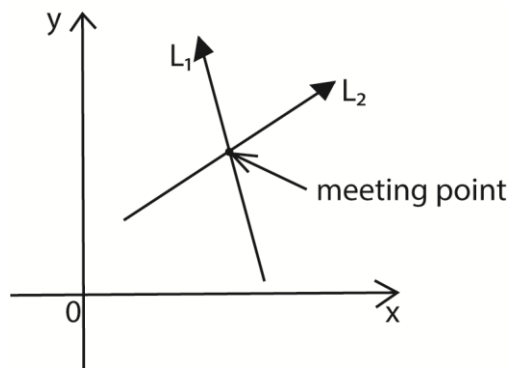
$$a = 3, b = -1 \text{ and } c = -1$$

Substituting for $(x, y) = (8, -5)$

$$d = \left| \frac{3(8) - 1(-5) - 1}{\sqrt{3^2 + (-1)^2}} \right| = \frac{28}{\sqrt{10}} = 8.85 \text{ units}$$

Intersection of two lines

The point of intersection of two or more lines is obtained by solving the equations of the lines simultaneously.



Example 9

- (a) Find the coordinates of the point of intersection of each of the following pairs of straight lines

- (i) $y = x + 3$ and $y = 4x + 6$

Solution

$$\text{Let } y = x + 3 \dots\dots\dots(1)$$

$$\text{And } y = 4x + 6 \dots\dots\dots(2)$$

Equating (1) and (2)

$$x + 3 = 4x + 6; x = -1$$

substituting x into (1)

$$y = 4(-1) + 6; y = 2$$

Hence the point of intersection is $(-1, 2)$

- (ii) $2x - 3y = 7$ and $3x - 7y = 13$

$$\text{Let } 2x - 3y = 7 \dots\dots\dots(1)$$

$$\text{And } 3x - 7y = 13 \dots\dots\dots(2)$$

Equating (1) and (2)

$$x + 3 = 4x + 6; x = -1$$

$$3\text{eqn. (1)} - 2\text{eqn. (2)}$$

$$5y = -5 \Rightarrow y = -1$$

Substituting for y into eqn. (1)

$$2x - 3(-1) = 7; x = 2$$

Hence the point of intersection $(2, -1)$

- (b) (i) Find the equation of the straight line L , which passes through the point $P(2, 4)$ and perpendicular to the line $5y + x = 7$

Solution

Let $Q(x, Y)$ lie on the same line

$$\text{Gradient of } PQ, m_1 = \frac{y-4}{x-2}$$

$$\text{From } 5y + x = 7; y = -\frac{1}{5}x + \frac{7}{5}$$

$$\text{Gradient, } m_2 = -\frac{1}{5}$$

But for perpendicular lines, $m_1 \times m_2 = -1$

$$\Rightarrow -\frac{1}{5} \left(\frac{y-4}{x-2} \right) = -1$$

$$y - 4 = 5(x - 2)$$

$$y = 5x - 6$$

- (ii) Given that the line L meets line $y = x + 6$ at point S. find the coordinates of S.

Solution

At S the two equations are equal

$$\Rightarrow 5x - 6 = x + 6; x = 3$$

Substituting x in $y = x + 6$

$$y = 3 + 6 = 9$$

Hence coordinates of S are (3, 9)

- (c) The line L has equation $2x - y - 1 = 0$. The line M passes through point A(0, 4) and is perpendicular to the line L. The line N passes through point B(3, 0) and is parallel to M.

- (i) Find an equation of M and show that the line L and M intersect at the point A(0, 4).

Solution

Let K(x, y) lie on line M

$$\text{Gradient of AK, } m_1 = \frac{y-4}{x-0} = \frac{y-4}{x}$$

$$\text{From } 2x - y = 1; y = 2x - 1$$

$$\text{Gradient, } m_2 = 2$$

For perpendicular line $m_1 \times m_2 = -1$

$$\Rightarrow 2 \left(\frac{y-4}{x} \right) = -1$$

$$2(y - 4) = -x$$

$$2y + x = 8$$

At the point, P of intersection, the two lines are equal

$$2x - 1 = y \dots\dots\dots(1)$$

$$2y + x = 8 \dots\dots\dots(2)$$

Substituting for y in eqn. (1) into equation (2)

$$2(2x - 1) + x = 8$$

$$x = 2$$

Substituting for x in eqn. (1)

$$y = 2(2) - 1 = 3$$

Hence point of intersection P (2, 3)

- (ii) Find an equation of N and hence find the coordinates of point Q where the Line L and Line N intersect

Let D (x, y) lie on N

$$\text{Gradient BD, } m_1 = \frac{y-0}{x-3} = \frac{y}{x-3}$$

From line $2y + x = 8$

$$y = -\frac{1}{2}x + 4$$

$$\text{Gradient } m_2 = -\frac{1}{2}$$

For parallel lines $m_1 = m_2$.

$$\Rightarrow -\frac{1}{2} = \frac{y}{x-3}$$

$$2y = 3 - x$$

$$2y + 3x = 3$$

At point Q, Line L = line N

$$2x - 1 = y \dots\dots\dots(1)$$

$$2y + x = 3 \dots\dots\dots(2)$$

Substituting for y in eqn. (2)

$$2(2x - 1) + x = 3$$

$$5x - 2 = 3$$

$$x = 1$$

Substituting for x in eqn. (1)

$$y = 2(1) - 1 = 1$$

hence Q(1, 1)

- (iii) Prove that $\overline{AP} = \overline{BQ} = \overline{PQ}$

$$\overline{AP} = \sqrt{(2-0)^2 + (3-4)^2} = \sqrt{5}$$

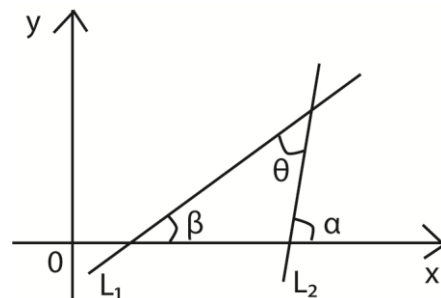
$$\overline{BQ} = \sqrt{(1-3)^2 + (1-0)^2} = \sqrt{5}$$

$$\overline{PQ} = \sqrt{(1-2)^2 + (1-3)^2} = \sqrt{5}$$

Hence $\overline{AP} = \overline{BQ} = \overline{PQ}$

Calculating angle between two lines

Suppose that two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are inclined at angle α and β respectively as shown below



The sum of two interior angles = opposite exterior angle

$$\Rightarrow \theta + \beta = \alpha$$

$$\begin{aligned}\theta &= \alpha - \beta \\ \tan \theta &= \tan (\alpha - \beta) \\ &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ &= \frac{m_1 - m_2}{1 + m_1 m_2} \\ \theta &= \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)\end{aligned}$$

Hence the acute angle between two intersecting lines is $\tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$

Note that the angle between parallel lines is zero while that between perpendicular line is 90° .

Example 10

- (a) Find the acute angle between the following pairs of lines

(i) $4y - 3x = 6$ and $2y + x = 3$

Solution

For $4y - 3x = 6$

$$y = \frac{3}{4}x + \frac{6}{4} \Rightarrow m_1 = \frac{3}{4}$$

For $2y + x = 3$

$$y = -\frac{1}{2}x + \frac{3}{2} \Rightarrow m_2 = -\frac{1}{2}$$

From $\theta = \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$

$$\begin{aligned}\theta &= \tan^{-1} \left(\frac{\frac{3}{4} - \left(-\frac{1}{2}\right)}{1 + \frac{3}{4} \left(-\frac{1}{2}\right)} \right) = \tan^{-1} \left(\frac{\frac{3}{4} + \frac{1}{2}}{1 - \frac{3}{8}} \right) \\ &= \tan^{-1} 2 = 63.4^\circ\end{aligned}$$

(ii) $2x - 5y = 15$ and $3y + 2x = 6$

Solution

For $2x - 5y = 15$

$$y = \frac{2}{5}x + 3 \Rightarrow m_1 = \frac{2}{5}$$

For $3y + 2x = 6$

$$y = -\frac{2}{3}x + 2 \Rightarrow m_2 = -\frac{2}{3}$$

From $\theta = \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$

$$\begin{aligned}\theta &= \tan^{-1} \left(\frac{\frac{2}{5} - \left(-\frac{2}{3}\right)}{1 + \frac{2}{5} \left(-\frac{2}{3}\right)} \right) = \tan^{-1} \left(\frac{\frac{2}{5} + \frac{2}{3}}{1 - \frac{4}{15}} \right) \\ &= \tan^{-1} \left(\frac{16}{11} \right) = 46.8^\circ\end{aligned}$$

(iii) $y = 4$ and $3y + 2x - 6 = 0$

Solution

For $y = 4$, $\Rightarrow m_1 = 0$

For $3y + 2x - 6 = 0$

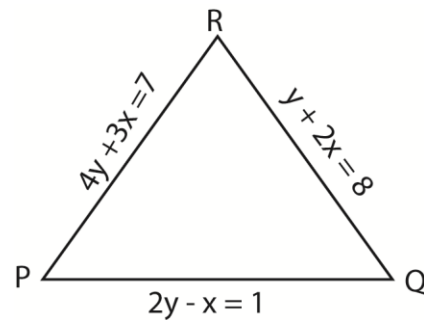
$$y = -\frac{2}{3}x + 2 \Rightarrow m_2 = -\frac{2}{3}$$

From $\theta = \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$

$$\begin{aligned}\theta &= \tan^{-1} \left(\frac{0 - \left(-\frac{2}{3}\right)}{1 + 0 \cdot \left(-\frac{2}{3}\right)} \right) = \tan^{-1} \left(\frac{\frac{2}{3}}{1} \right) \\ &= \tan^{-1} \left(\frac{2}{3} \right) = 33.7^\circ\end{aligned}$$

- (b) Calculate the area of the triangle which has sides given by the equations $2y - x = 1$, $y + 2x = 8$ and $4y + 3x = 7$

Solution



At point P

$$4y + 3x = 7 \dots\dots\dots(1)$$

$$2y - x = 1 \dots\dots\dots(2)$$

$$\text{Eqn. (1)} - 2\text{eqn. (2)}$$

$$10y = 10; y = 1$$

$$\text{Substituting for } y \text{ in eqn. (2)}$$

$$x = 2(1) - 1 = 1$$

$$\text{Hence } P(1, 1)$$

At point Q

$$y + 2x = 8 \dots\dots\dots(1)$$

$$2y - x = 1 \dots\dots\dots(2)$$

$$\text{Eqn. (1)} + 2\text{eqn. (2)}$$

$$5y = 10; y = 2$$

$$\text{Substituting for } y \text{ in eqn. (2)}$$

$$x = 2(2) - 1 = 3$$

$$\text{Hence } Q(3, 2)$$

At point R

$$y + 2x = 8 \dots\dots\dots(1)$$

$$4y + 3x = 7 \dots\dots\dots(2)$$

$$3\text{Eqn. (1)} - 2\text{eqn. (2)}$$

$$-5y = 10; y = -2$$

Substituting for y in eqn. (1)

$$2x = 8 + 2 = 10 \Rightarrow x = 5$$

Hence r(-2, 5)

Finding dimensions

$$\overline{PR} = \sqrt{(5-1)^2 + (-2-1)^2} = 5$$

$$\overline{PQ} = \sqrt{(3-1)^2 + (2-1)^2} = \sqrt{5}$$

$$\overline{QR} = \sqrt{(5-3)^2 + (-2-2)^2} = 2\sqrt{5}$$

Finding $\angle QPR$

$$\text{For } 4y + 3x = 7$$

$$y = -\frac{3}{4}x + \frac{7}{4} \Rightarrow m_1 = -\frac{3}{4}$$

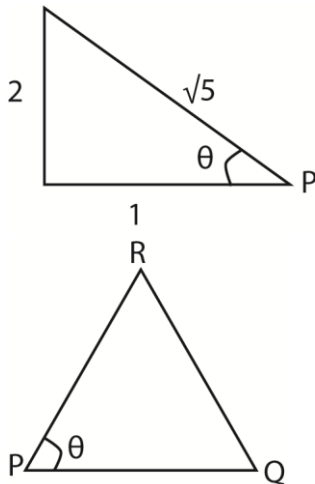
$$\text{For } 2y - x = 1$$

$$y = \frac{1}{2}x + \frac{1}{2} \Rightarrow m_2 = \frac{1}{2}$$

$$\text{From } \theta = \tan^{-1} \left(\frac{m_1 - m_2}{1 + m_1 m_2} \right)$$

$$\angle QPR = \tan^{-1} \left(\frac{\frac{1}{2} - (-\frac{3}{4})}{1 + \frac{1}{2}(-\frac{3}{4})} \right) = \tan^{-1} \left(\frac{\frac{1+3}{2}}{1-\frac{3}{8}} \right)$$

$$= \tan^{-1}(2)$$



Method 1 using sine rule

$$\begin{aligned} \text{Area of PQR} &= \frac{1}{2} \times PQ \times PR \sin \theta \\ &= \frac{1}{2} \times \sqrt{5} \times 5 \times \frac{2}{\sqrt{5}} = 5 \text{ sq. units} \end{aligned}$$

Method 2 using Heron's formula

$$\text{Area of PQR} = \sqrt{s(s-a)(s-b)(s-c)}$$

Where a, b, and c are sides of a triangle and

$$s = \frac{1}{2}(a + b + c)$$

$$s = \frac{1}{2}(5 + \sqrt{5} + 2\sqrt{5}) = \frac{5+3\sqrt{5}}{2}$$

$$\text{Area} = \sqrt{\left(\frac{5+3\sqrt{5}}{2}\right)\left(\frac{3\sqrt{5}-5}{2}\right)\left(\frac{5+\sqrt{5}}{2}\right)\left(\frac{5-\sqrt{5}}{2}\right)}$$

$$= 5 \text{ sq. units}$$

Revision Exercise 3

- Find the gradient of each of the following straight lines
 - $y = 4x - 2$ [4]
 - $y = 2x + 3$ [3]
 - $y = 2 - 5x$ [-5]
 - $\frac{y}{2} - \frac{x}{5} = 4$ $\left[\frac{2}{5}\right]$
- Find the equation of the straight line that has the following properties
 - Gradient 1 and passes through (2, 4)
[$y = x + 2$]
 - gradient $\frac{1}{4}$ and passes through (2, 5)
[$y = \frac{1}{4}x + \frac{9}{2}$]
- Find the equation of a straight line that has the following properties
 - Passes through (-2, 3) and parallel to $y = 5x + 4$
[$y = 5x + 13$]
 - Passes through (6, -2) and is perpendicular to $y = -3x + 4$
[$y = \frac{1}{3}x - 4$]
 - Passes through $\left(-\frac{1}{3}, -\frac{2}{3}\right)$ and is perpendicular to $3y + 10x - 8 = 0$
[$10y = 3x - 3$]
- Find the equation of a straight line joining the following pairs of points
 - (2, 4) and (-1, 0) [$3y = 4x + 4$]
 - (-4, 1) and (6, 2) [$10y = x + 14$]
 - (3, 4) and (-1, 4) [$y = 4$]
- Find the equation of the perpendicular bisector of the straight lines joining each of the following pairs of point
 - (5, 5) and (2, -2) [$4y + 2x = 11$]
 - (-1, 4) and (3, 3) [$2y - 8x + 1 = 0$]
 - (3, 2) and (-4, 1) [$y = -7x - 2$]
- Find the equation of a straight line:
 - L_1 which is perpendicular bisector of points A(-2, 3) and B(1, -5)
[$16y - 6x + 13 = 0$]

- (b) L_2 which is a perpendicular bisector of the points $B(1, -5)$ and $(17, 1)$
 $[3y + 8x - 60 = 0]$
- (c) Show that L_1 is perpendicular to L_2 .
7. The perpendicular bisector of a straight line joining the points $(3, 2)$ and $(5, 6)$ meet the x-axis at A and they-axis at B. Prove that the distance AB is equal to $\sqrt{5}$.
8. A is a point $(1, 2)$ and B is a point $(7, 4)$. The straight line L_1 passes through B and is perpendicular to AB; the straight line L_2 passes through A and is also perpendicular to AB. The line L_1 meets the x-axis at P and the y-axis at Q. Line L_2 meets the x-axis at R and the y-axis at S.
- (a) Find the equations of each of L_1 and L_2 .
 $[y = -3x + 25, y = -3x + 5]$
- (b) Calculate the area of the triangle OPQ.
 $\left[104\frac{1}{6}\right]$
- (c) Calculate the area of the triangle ORS.
 $\left[4\frac{1}{6}\right]$
- (d) Find the area of the trapezium PQSR.
 $[100 \text{ sq. units}]$
9. P is the point with coordinates $(2, 1)$ and L is the straight line which is perpendicular to OP and which passes through P.
- (a) Find the equation of L. $[y = -2x + 5]$
- (b) Given that line L meets the x-axis at A and y-axis at B. calculate
- (i) the area of the triangle OAP. $[1.25]$
- (ii) The area of the triangle OBP $[5]$
- (iii) Find the ratio of the area OAP to that of OBP $[1:4]$
10. Find the shortest distance between each of the following
- (a) The point $(2, 4)$ and the line
 $3x - 4y + 8 = 0$ $\left[\frac{2}{5}\right]$
- (b) The point $(5, -1)$ and the line
 $12x + 5y - 3 = 0$ $[4]$
- (c) The point $(9, -3)$ and the line $y = x$.
 $[6\sqrt{2}]$
11. Find the coordinates of the point of intersection of each of the following pairs of straight lines
- (a) $y = 2x + 3$ and $y = 4x + 1$ $[1, 5]$
- (b) $y = x + 3$ and $y = 4x + 6$ $[-1, 2]$
- (c) $2x - 3y = 7$ and $3x - 7y = 13$ $[2, -1]$
- (d) $x + 3y - 2 = 0$ and $3x + 5y - 8 = 0$ $\left[\frac{7}{2}, -\frac{1}{2}\right]$
12. Find the equation of the straight line L, which passes through the point $(2, 4)$ and perpendicular to the line $5y + x = 7$
 $[y = 5x - 6]$
- (b) Given that the line L meets the line $y = x + 6$ at point S, find the coordinates of point S. $[3, 9]$
13. Calculate the area of the triangle which has sides given by the equations $2y - x = 1$, $y + 2x = 8$ and $4y + 3x = 7$ $[5 \text{sq. units}]$
14. The point A has coordinates $(2, -5)$. The straight line $3x + 4y - 36 = 0$ cuts the x-axis at B and the y-axis at C. Find
- (a) The equation of the line through A which is perpendicular to the line BC.
 $[4x - 3y = 23]$
- (b) The perpendicular distance from the line BC. $[10]$
- (c) The area of the triangle ABC. $[75 \text{ sq. units}]$

Locus

A locus is the set of all points in a plane that satisfy some condition. For example the locus of points equidistant from two given points say A and B is a perpendicular bisector of AB.

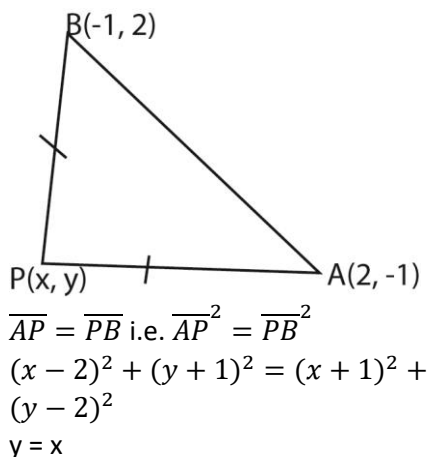
Locus can be expressed in terms of the Cartesian coordinates (x, y) of the form (r, θ)

Example 11

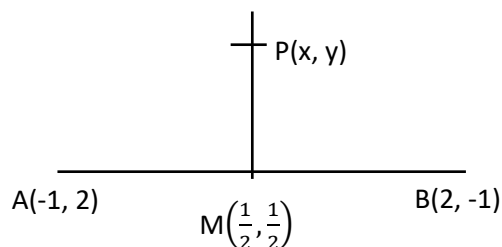
- (i) Find the locus of point $P(x, y)$ that is equidistant from the point $A(2, -1)$ and $B(-1, 2)$

Solution

Method 1



Method 2



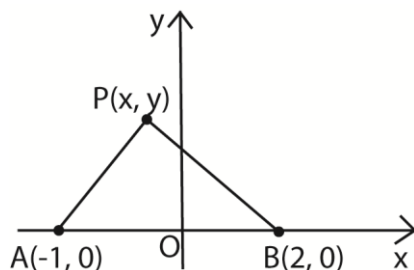
$$\text{Gradient of } AB = \frac{2+1}{-1-2} = -1$$

Since AB and MP are perpendicular, the product of their gradients = -1

$$\text{Gradient of } MP = \frac{x - \frac{1}{2}}{y - \frac{1}{2}} = 1$$

$$y = x$$

- (ii) Find the locus of a point which moves so that of the squares of its distance from points A(-2, 0) and B(2, 0) is 25 units.



$$\overline{AP}^2 + \overline{BP}^2 = 25$$

$$(x+1)^2 + y^2 + (x-2)^2 + y^2 = 25$$

$$x^2 + y^2 = 25$$

- (iii) The locus of P(x, y) is such that the distance OP is half the distance PR, where O is the origin and R is the point (-3, 6)

- (i) Show that the locus of P describes a circle in the x-y plane

Solution

$$\text{Given } OP = \frac{1}{2} PR \text{ i.e. } 4OP^2 = PR^2$$

$$4(x^2 + y^2) = (x+3)^2 + (y-6)^2$$

$$4x^2 + 4y^2 = x^2 + 6x + 9 + y^2 - 12y + 36$$

$$x^2 + y^2 - 2x + 4y - 15 = 0 \text{ (a circle in } x-y \text{ plane)}$$

- (ii) Determine the radius of the circle and the centre of the circle.

Solution

$$x^2 + y^2 - 2x + 4y = 15$$

After completing squares we have

$$(x-1)^2 + (y+2)^2 = 20$$

The centre of the circle is (1, -2) and the radius = $\sqrt{20} = 2\sqrt{5}$ units

- (iii) Where does P cut the line $x = 3$?

Solution

Substituting $x = 3$ in the equation

$$(x-1)^2 + (y+2)^2 = 20$$

$$((3-1)^2 + (y+2)^2 = 20$$

$$(y+2)^2 = 16 \text{ i.e. } y+2 = \pm 4$$

$$y = -2 \pm 4 \text{ i.e. } y = -6 \text{ or } y = 2$$

$\therefore$ P cuts the line $x = 3$ at the point (3, 2) and (3, -3)

- (iv) Find the locus of P(x, y) if its distance from A(-1, 1) is equal to the distance from the line $2x - y = 1$

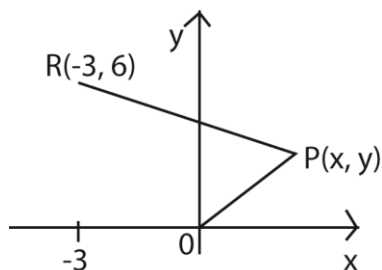
Solution

$$\begin{aligned}
 2\sqrt{(x+2)^2 + y^2} &= \\
 3\sqrt{(x-8)^2 + (y-6)^2} & \\
 \text{Squaring both sides} & \\
 4(x^2 + 4x + 4 + y^2) &= \\
 = 9(x^2 - 16x + 64 + y^2 - 12y + 36) & \\
 4x^2 + 16x + 16 + 4y^2 &= \\
 = 9x^2 - 144x + 900 + 9y^2 - 108y & \\
 5x^2 + 5y^2 - 160x - 108y + 884 = 0 &
 \end{aligned}$$

Example 12

The locus of P is such that the distance of OP is half distance PR, where O is the origin and R is the point (-3, 6)

- (i) Show that the locus of P describes a circle in the x-y plane



$$\begin{aligned}
 PR &= 2OP \\
 (PR)^2 &= 4(OP)^2 \\
 \text{But } (OP)^2 &= (x-0)^2 + (y-0)^2 \\
 &= x^2 + y^2 \\
 (PR)^2 &= (x+3)^2 + (y-6)^2 \\
 \Rightarrow (x+3)^2 + (y-6)^2 &= 4(x^2 + y^2)
 \end{aligned}$$

Applications of coordinate geometry 1

1. Navigation and GPS

Application: Determining the shortest path or distance between two locations.

Example: GPS systems use coordinates (latitude and longitude) to calculate the distance between your current location and your destination using the distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

simplifying

$$x^2 + y^2 - 2x + 4y - 15 = 0$$

Hence the locus of P describes a circle in x-y plane

- (ii) Determine the radius and the center of the circle

$$\text{Comparing } x^2 + y^2 - 2x + 4y - 15 = 0 \text{ with } x^2 + y^2 + 2gx + 2fy + C = 0$$

$$g = -1, f = 2, c = -15$$

$$\text{Center of the circle is } (-g, -f) = (1, -2)$$

$$\begin{aligned}
 \text{Radius } R &= \sqrt{g^2 + f^2 - c} \\
 &= \sqrt{(-1)^2 + (2)^2 - (-15)} \\
 &= \sqrt{1 + 4 + 15} \\
 &= \sqrt{20} \\
 &= 4.4721(4d.p)
 \end{aligned}$$

- (iii) Where does P cut the line $x = 3$?

Substituting 3 in the equation

$$x^2 + y^2 - 2x + 4y - 15 = 0$$

$$y = -6 \text{ or } 2$$

Hence P cuts the line $x = 3$ at (3, 2) and (3, -6)

2. Architecture and Engineering

Application: Designing buildings, bridges, and layouts with precision.

Example: Architects use coordinate geometry to plot blueprints and ensure accurate placement of structural elements using points, lines, and angles on a coordinate plane.

3. Computer Graphics and Animation

Application: Creating and manipulating digital images and animations.

Example: Video games and animation software use coordinate geometry to move characters, detect collisions, and render 3D environments.

4. Physics and Kinematics

Application: Analyzing motion and forces.

Example: The trajectory of a projectile or the path of a moving object can be modeled using equations of lines and curves in coordinate space.

5. Robotics and Automation

Application: Programming movement and spatial awareness.

Example: Robots use coordinate geometry to navigate environments, avoid obstacles, and reach target positions using coordinate-based path planning.

6. Astronomy

Application: Mapping stars and planetary motion.

Example: Astronomers use coordinate systems to track celestial bodies and predict their positions over time.

Revision Exercise 4

- Find the equation to the locus of a point which moves so that its distance from the point (3, 4) is always 5 units.
 $[x^2 + y^2 - 6x - 8y = 0]$
- A and B are points (1, 0) and (7, 8) respectively. A point P moves so that the angle APB is a right angle. Find the equation of locus of P.
 $[x^2 + y^2 - 4x - 8y + 20 = 0]$
- A variable point P is given by parametric equation $x = ct$, $y = \frac{c}{t}$. Show that the locus of P is $xy = c^2$.
- Find the locus of point P which is equidistant from the line $x = 2$ and the circle $x^2 + y^2 = 1$.
 $[x^2 + 6x - 9 = 0]$
- A Point P is twice as far from the line $x + y = 5$ and from the point (3, 0). Find the locus of P.
 $[7x^2 + 7y^2 - 38x + 10y - 2xy + 47 = 0]$
- Find the Cartesian equation of a curve given parametrically by

(i) $x = \frac{1+t}{1-t}$ and $y = \frac{2t}{1-t}$ [$y - x + 1 = 0$]

(ii) $x = 2t$ and $y = \frac{2}{t} + 1$ and sketch it
 $y = \frac{4+x}{x}$

- (iii) P(3, 2) and Q(5, 6) are two fixed points and R moves so that angle PRQ is right angle. Show that the locus of R is given by the equation $x^2 + y^2 - 9x - 6y + 26 = 0$

- (iv) Show that the locus of P is $\frac{1}{4}x^2 + \frac{1}{3}y^2 = 1$, given that $PA + PB = 4$ where A and B are the points (1, 0) and (-1, 0) respectively

Exercise 5 (Topical question)

- ABCD is a quadrilateral with A(2, -2), B(5, -1), C(6, 2) and D(3, 1). Show that the quadrilateral is a rhombus.
- PQRS is a quadrilateral with vertices P(2, -1), Q(4, -1) and S(2, 1). Show that the quadrilateral is a rhombus
- The Locus of P is such that the distance OP is half the distance PR, where O is the origin and R is the point (-3, 6).

- (a) Show that the locus of P describes a circle in $x - y$ plane
 $[x^2 + y^2 - 2x + 4y - 15 = 0$ (a circle in $x - y$ plane)]
- (b) Determine the centre and radius of the circle.
 [The centre of the circle is (1, -2) and the radius = $\sqrt{20} = 2\sqrt{5}$ units]
- (c) Where does P cut the line $x = 3$
 [(3, 2) and (3, -3)]
4. A Point P is twice as far from the line $x + y = 5$ as from the point (3, 0). Find the locus of P.
 $[7x^2 + 7y^2 - 38x + 10y + 47 = 0]$
5. Find the locus of point P which is equidistant from the line $x = 2$ and the circle $x^2 + y^2 = 1$.
 $[x^2 + 6x - 9 = 0]$
6. The points R(2, 0) and P(3, 0) lie on the x -axis and Q(0, -y) lie on the y -axis. The perpendicular from the origin to RQ meets PQ at S(X, -Y). Determine the locus of S in terms of X and Y. $[2X^2 + 3Y^2 - 6X = 0]$
7. The point A(2, 1), P(α , β) and point B(1, 2) lie in the same plane. PA meets the x -axis at point (h, 0) and PB meets the y -axis at point (0, k). Find h and k in terms of α and β .
 $\left[h = \frac{2\beta - \alpha}{\beta - 1}; k = \frac{2\alpha - \beta}{\alpha - 1}\right]$
8. A is a point (1, 3) and B is a point (4, 6). P is a variable point which moves in such a way that $\overline{AP}^2 + \overline{PB}^2 = 34$. Show that the locus of P describes a circle. Find the centre and radius of the circle
 $\left[\text{centre is } \left(\frac{5}{2}, \frac{9}{2}\right), \text{radius} = \frac{2\sqrt{5}}{2} \text{ units}\right]$
9. Find the locus of the point P(x, y) which moves such that its distance from the point S(-3, 0) is equal to its distance from a fixed line $x = 3$ $[y^2 + 12x = 0]$
10. Given the vector $a = i - 3j + 3k$ and $b = -i - 3j + 2k$. find
- (iv) acute angle between vectors a and b
 $[30.86^\circ]$
- (ii) equation of the plane containing a and b $[-3x + 5y + 6z = 0]$
11. The points A and B lie on the positive sides of the x -axis and y -axis respectively. If the length of AB is 5 units and angle OAB is θ , where O is the origin, find the equation of the line AB (leave θ in your answer)
 $[y = -x \tan \theta + 5 \sin \theta]$
12. (a) Find the equation of the locus of a point which moves such that its distance from D(4, 5) is twice its distance from the origin.
 $[8x^2 + 8y^2 + 8x + 10y - 41 = 0]$
 (b) the line $y = mx$ intersects the curve $2x^2 - x$ at point A and B. Find the equation of the locus of point P which divides AB in the ratio 2:5. $[y = 7x^2 - x]$
13. Find the locus of the point which is equidistant from the line $x = 2$ and the circle $x^2 + y^2 = 1$ $[y^2 + 6x - 9 = 0]$
14. The point R(2, 0) and P(0, 0) lie on x -axis and Q(0, -y) lie on the y -axis. The perpendicular from the origin to RQ meets PQ at point (X, -Y). Determine the locus of S in terms of X and Y $[2X^2 + 3Y^2 - 6X = 0]$
15. The point A(2, 1), P(α , β) and point B(1, 2) lie in the same plane, PA meets the x -axis at point (h, 0) and PB meets the y -axis at the point (0, k). Find h and k in terms of α and β . $\left[h = \frac{2\beta - \alpha}{\beta - 1}, k = \frac{2\alpha - \beta}{\alpha - 1}\right]$
16. A is a point (1, 3) and B the point (4, 6), P is a variable point which moves in such a way that $AP^2 + PB^2 = 34$. Show that the locus of P describes a circle. Find the center and radius of the circle. $\left[\left(\frac{5}{2}, \frac{9}{2}\right), \frac{5\sqrt{2}}{2}\right]$

Thank you Dr. Bbosa Science